

COURSE NAME:

SIMATS ENGINEERING ASSESSMENT
TOOL 2: Concept Mapping

Cloud Computing and COURSE CODE: Big
Data Analytics

CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)

Question Count: 4

Total Marks: 20

Code of Conduct

I Logeshwaran R (19251145) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Lohn
R.

Assessment Tasks

Concept Mapping

Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy. 1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations. 2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution,

distributed systems, and service delivery.

3. Concept Map 3: **Service Models**. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user

responsibilities.

4. Concept Map 4: Virtualization and Web **Services as Enablers** of Cloud. Show how these two concepts support

cloud deployment and scalability.

Expected **concept** nodes:

Elasticity, scalability, resource pooling, on-demand **access**, measured service

Virtual machine, hypervisor, abstraction, **API**, SOAP, REST, provider, consumer
Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria

Weightage

Concept

25%

Identification

Level 4 - Exemplary All major concepts are accurately identified

Level 3- Proficient

Level 2 - Developing

are identified

Most concepts are Limited concepts identified correctly

Level 1 - Beginning
Essential concepts are missing

Relationship Mapping

25%

Links between concepts are accurate and

Most links are correct



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Cloud Computing and Big Data

1. Concept Map: CLOUD COMPUTING FUNDAMENTALS.

Cloud computing

cloud computing is the delivery of computing

Services over the internet on-demand, with

Characteristics

pay

as-you-go pricing.

Benefits

*On demand

*cost savings

Scalability

self service.

* Broad

network access

* Resource

Spooting

.

*

Rapid
elasticity

* measured

service

Apexibility

* High

availability

* Reliability

* access anywhere

limitations.

* Internet dependency * security and privacy

poncerns

vendor lockin

Downtime/ outages

* performance

variability,

CLOUD

COMPUTING

2. Concept map: EVOLUTION OF CLOUD COMPUTING.

Hardware

evolution

- * mainframes

- * **personal computer**

- *

personal

server

internet

software

evolution

- * **web** applications

Distributed

computing

.

system

ems

- * Grid

com

puti

ng.

Internet

evolution,

- * LAN

- * WAN

- * web services.

- * Cluster

- * 80A

- *

High Speed

>computing.

- * utility

computing

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Moder cloud platforms . .

*aws

* Microsoft azure * google cloud.

1

3. Concept map:

Shard

responsibility

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Appl

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on/
layer

Plat
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layer

Infra
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[Ph
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CLOUD
SERVILE MODELS

Cloud service

models

laas

Paas

*provid
es

visited
computi
ng over
luted

Data
Cent
er
*serv
ers
#stor
age

#networking.

* provide
platform for
developing,
testing

* Data entry

.

*

Servers

OS

*

* networking * Runtime

* Middleware

SaaS

Delivers ready to use software.

*

* Data center

Storage

OS

#middleware

application

XaaS

* Any IT
service,
delivered

our

the
cloud

*Deli
ver
spec
ific

servic

Amazon

GGoogle app

microsoft.

Doop'

box

aws

engine

365

4. Concept map: VIRTUALIZATION & WEB SERVICES

Virtualization

AS ENABLE's

web services

Technologies that enable

Technology that (nates

by

attracting
pleyical

communication and interaction

resounes

key compon ents.

puer

the internet

-key
components

Hypervisor

VM

Ryource

API

SOAP

PEST

Porting

Crats and

Run

Pu,

Define

XML

manage

mutiply

memory
memory,

muthood

Bend

01

storage

protocol

lightweight
protoc

"How they chable

Lond)

RESULT

* virtualization enables
efficient
resource utilization.

* web services enable stand
and

communication.

* Together they support on-
demand provisioning.

Flexible

Deployment

High

scalable

cost efficient

global

acurbiliz

support cloud development tools.

Public cloud

Private
cloud

Hybrid
Hoid

Communit
y
Cloud